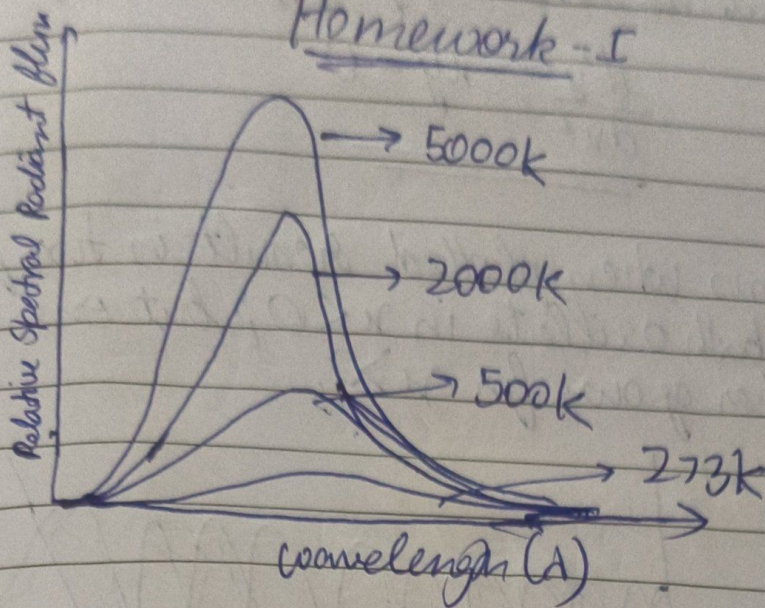


Characterization of Materials-I

Homework-I

From Wein's Displacement Law:-

$$\lambda_{\text{max}} T = b$$

$$\lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{T}$$

$$\lambda_1 = 5.796 \times 10^{-7} \text{ m} = 579.6 \text{ nm} = 17253.2 \text{ cm}^{-1}$$

$$\lambda_2 = 1.449 \times 10^{-7} \text{ m} = 144.9 \text{ nm} = 6901.3 \text{ cm}^{-1}$$

$$\lambda_3 = 5.796 \times 10^{-6} \text{ m} = 5796 \text{ nm} = 1725.3 \text{ cm}^{-1}$$

$$\lambda_4 = 1.061 \times 10^{-5} \text{ m} = 10615 \text{ nm} = 942 \text{ cm}^{-1}$$

IR spectrum typically lies b/w 400-4000 cm^{-1} .

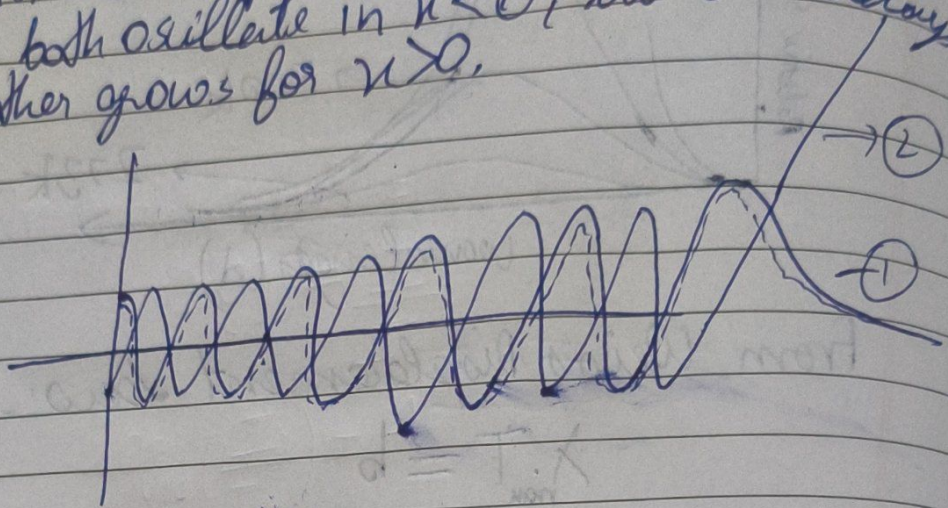
∴ Therefore λ_3 & λ_4
 $\lambda_3 = 5796 \text{ nm}$ is preferred for IR spectroscopy.

∴ Radiation at ~~5000K~~ ^{500K} is used.

3) Airy functions are special functions that arise as solutions to the Airy differential eqn.

$$\frac{d^2 y}{dx^2} - xy = 0$$

Solutions when plotted result in two functions which both oscillate in $x < 0$, but one decays and other grows for $x > 0$.



4) Diffraction grating: An optical element that separates polychromatic light into its constituent wavelengths (colors).
It consists of equally spaced parallel grooves formed on a reflective coating deposited on a substrate.

→ When light is incident on the grating, each slit acts as a source of diffracted waves which interfere.

→ Different wavelengths constructively interfere at different angles, causing spatial separation of colours.

→ The physical requirements for a ~~good~~ grating is that there must be sufficient no. of grooves per millimetre to provide adequate dispersion.

→ Also width of grooves must be of order of wavelength of incident light.

$$\psi_1 = A e^{ikx}$$

$$\psi_2 = A e^{i(kx+\delta)}$$

$$\psi_1 + \psi_2 = A [e^{ikx} + e^{i(kx+\delta)}]$$

$$= A [\cos kx + \cos(kx+\delta) + i(\sin kx + \sin(kx+\delta))]$$

$$= A \left[2 \cos\left(kx + \frac{\delta}{2}\right) \cos \frac{\delta}{2} + i 2 \sin\left(kx + \frac{\delta}{2}\right) \cos \frac{\delta}{2} \right]$$

$$\psi_1 + \psi_2 = \frac{2A \cos \delta}{2} A e^{i(kx+\delta/2)}$$

b) a) σ MO :- bond along joining axis.

→ stronger bonds.

→ Bonding σ MO has no nodal planes.

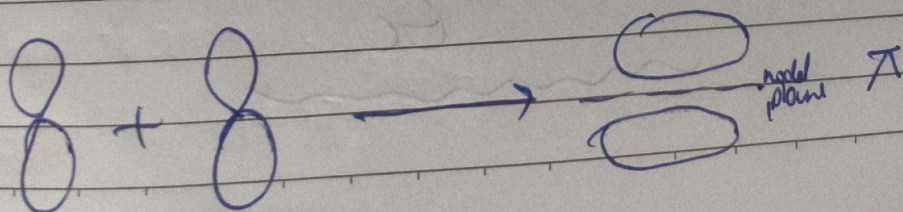
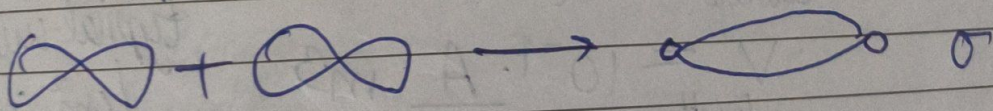
→ Both same phase orbitals combine to form a bonding σ MO.

π MO :- bond parallel to joining axis.

→ Bonding π MO has 1 nodal plane.

→ out of phase orbitals combine to form a bonding π MO.

→ weaker than σ MO.



b) Bonding MO are stable and favourable
→ have lesser energy than non-bonding orbitals

7) 1% change in intensity =

$$0.99 = \frac{I}{I_0}$$

$$\log_{10}\left(\frac{I_0}{I}\right) = A$$

~~A = 0.004364~~ $A = 0.004364$

From Beer-Lambert Law:

$$A = \epsilon c l$$

~~2 = \epsilon c l~~ $0.004364 = \epsilon c l$

~~$l = \frac{2}{\epsilon c}$~~

$$l = \frac{A}{\epsilon c}$$

$$V_{\text{cuvette}} = A_c \cdot l$$

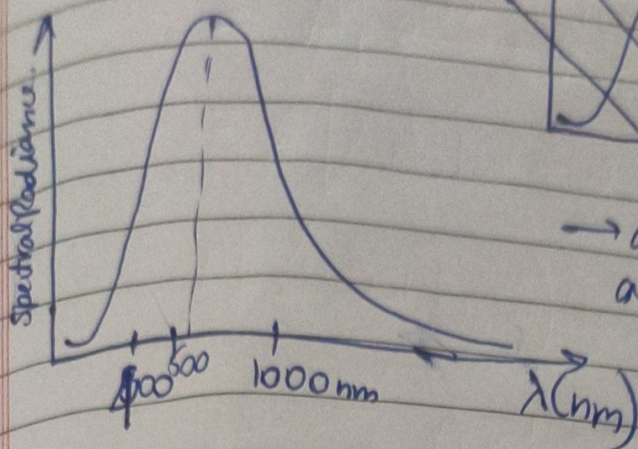
$$V_{\text{cuvette}} = 10^{-4} \cdot \frac{A}{\epsilon c} \text{ m}^3$$

typical cuvette has
1 cm² cross section



1) Approximate surface temperature of Sun = 5778K

Planck's Radiation Curve:



→ depicts Sun's emission as a Black Body.

→ peaking in visible range of 500 nm corresponding to green light.

• The AM1.5 spectrum represents sunlight reaching Earth's surface after passing through 1.5 times the thickness of Earth's atmosphere.

→ The AM1.5 spectrum has missing or reduced intensity at certain wavelengths due to atmospheric absorption and scattering.

• Atmospheric absorption:- O_3 absorbs UV at $\lambda < 300 \text{ nm}$

:- H_2O absorbs IR at $\lambda = 900, 1100 \text{ nm}$

CO_2 absorbs IR bands.

• Shorter wavelengths (blue & violet light) are scattered more ~~are~~ effectively by air molecules (Rayleigh scattering) giving the sky blue color and reducing direct solar irradiance at these wavelengths.